

Zsigmondy First Returns and Their Exact Prime-Power Congruence Tower

Route-A structural certificate C179 — round 1

Abstract

For coprime integers $a > b \geq 1$, multiplication by ab^{-1} on the finite unit group $(\mathbb{Z}/N\mathbb{Z})^\times$ defines an arithmetic dynamical system without fitted data. A prime p is a primitive divisor of $a^n - b^n$ exactly when the marked point 1 first returns at time n on the p -fiber. Classical Zsigmondy existence supplies a return prime except for its two exact exception forms; that theorem is attributed, not claimed new. If $e = v_p(a^n - b^n)$, we prove that the least return on every p^k -fiber is $np^{\max(0, k-e)}$. More generally, every admissible finite fiber is completely classified: all cycles have the multiplier order, its Artin–Mazur factor is reciprocal to the finite Koopman determinant, and inversion reverses time.

Keywords: primitive divisor; first return; congruence dynamics; multiplicative order; Zsigmondy theorem.

中文摘要

对互素整数 $a > b \geq 1$ ，在有限单位群 $(\mathbb{Z}/N\mathbb{Z})^\times$ 上乘以 ab^{-1} 给出一个无需拟合数据的算术动力系统。本文证明：素数 p 是 $a^n - b^n$ 的本原素因子，当且仅当标记点 1 在模 p 纤维上首次于时刻 n 返回。经典的 Zsigmondy 定理给出除两个精确例外类型之外的存在性；本文明确归属该经典结果，并不声称其为新定理。若 $e = v_p(a^n - b^n)$ ，则模 p^k 纤维上的最小返回时间精确等于 $np^{\max(0, k-e)}$ 。更一般地，本文完整分类 每个允许的有限纤维：所有循环具有相同长度，动力 ζ 因子与有限 Koopman 行列式互为倒数，取逆映射反转时间。

1 Frozen arithmetic dynamics

Fix coprime integers $a > b \geq 1$. For every $N \geq 2$ with $(N, ab) = 1$, set

$$U_N = (\mathbb{Z}/N\mathbb{Z})^\times, \quad q_N = ab^{-1} \pmod{N}, \quad R_N(x) = q_N x,$$

and mark $1 \in U_N$. One multiplication is one discrete source step. No logarithmic prime roof, prime weight, target table, or fitted parameter is introduced.

2 Primitive-return theorem

Theorem. For every prime p and $n \geq 2$, the following are equivalent:

1. $p \mid a^n - b^n$ and $p \nmid a^m - b^m$ for $1 \leq m < n$;
2. $p \nmid ab$ and $\text{ord}_p(ab^{-1}) = n$;
3. 1 has least positive return n under R_p .

Moreover, such a prime exists except when $(a, b, n) = (2, 1, 6)$, or when $n = 2$ and $a + b$ is a power of two.

Proof. A prime divisor of $a^n - b^n$ cannot divide ab . Thus $a^m \equiv b^m \pmod{p}$ is equivalent to $(ab^{-1})^m \equiv 1 \pmod{p}$. Excluding all earlier m is exactly the definition of multiplicative order, and the orbit of the mark is $1, q_p, q_p^2, \dots$. The existence and exception sentence is the classical Zsigmondy theorem [1]; no new proof or strengthening of that theorem is claimed. $\square$

Every primitive return prime for $n \geq 2$ is odd. Indeed, if a, b are odd, then $2 \mid a - b$ already at time one; if they have opposite parity, every $a^n - b^n$ is odd.

3 Exact lift through every prime power

Let p be primitive at n , and set $e = v_p(a^n - b^n)$. Then p is odd and $n = \text{ord}_p(ab^{-1}) \mid p - 1$. For every $k \geq 1$,

$$\boxed{\text{ord}_{p^k}(ab^{-1}) = np^{\max(0, k-e)}}. \quad (1)$$

Indeed, an exponent returning modulo p^k must be nr . Since b is a p -adic unit, odd-prime lifting of the exponent gives

$$v_p((ab^{-1})^{nr} - 1) = e + v_p(r).$$

The least eligible r is $p^{\max(0, k-e)}$. Thus every orbit on the p^k -fiber has the length in (1), and the number of cycles is $\varphi(p^k)/(np^{\max(0, k-e)})$.

4 Complete dynamics on every finite fiber

For any admissible N , write $L_N = \text{ord}_N(q_N)$. Translation returns a point x exactly when $q_N^t = 1$, independently of x . Hence U_N is $\varphi(N)/L_N$ cycles, each of length L_N , and

$$\# \text{Fix}(R_N^t) = \begin{cases} \varphi(N), & L_N \mid t, \\ 0, & L_N \nmid t, \end{cases} \quad \zeta_N(z) = (1 - z^{L_N})^{-\varphi(N)/L_N}. \quad (2)$$

For the finite permutation Koopman matrix $\mathcal{U}_N$,

$$\det(I - z\mathcal{U}_N) = (1 - z^{L_N})^{\varphi(N)/L_N}. \quad (3)$$

Inversion $J_N(x) = x^{-1}$ satisfies $J_N R_N J_N = R_N^{-1}$ because $(q_N x^{-1})^{-1} = x q_N^{-1}$.

5 Evidence and provisional Route-A boundary

The deterministic ledger contains 567 primitive-divisor rows, 2,080 prime-power lifts, and 1,650 complete finite fibers. A producer-independent checker passes 320,291 assertions; SymPy passes 6,674 exact checks; byte replay is exact; 60 repaired-hash mutations and one stale-hash mutation are all rejected. These are regression sentinels only; (1)–(3) have no cutoff.

The prime first-return relation is intrinsic arithmetic origin and therefore improves on a nonarithmetic control. It remains weak: a prime is a return *modulus*, not yet a selected global primitive orbit, and the source clock is not $\log p$. No target divisor, functional equation, counting law, or Hilbert–Pólya operator is claimed.

Scope. No target zero or prime table, local arithmetic factor, root number, automorphy input, prime-weighted global product, logarithmic prime roof, or Route-B authorization enters this note. Scope literal: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

No human, animal, clinical, personal, private, or sensitive data are used. No external funding or conflict of interest is reported. Package-local code and exact evidence accompany the paper. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer.

References

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